

9.2 – Disadvantages

While analysing electricity consumption patterns provides valuable insights, the project also faces several limitations and challenges. One major disadvantage is the availability and accuracy of data. Incomplete, outdated, or inconsistent electricity consumption records can affect the reliability of the analysis and lead to inaccurate conclusions.

Another challenge is the complexity of consumption behaviour. Electricity usage varies significantly based on factors such as weather conditions, household size, lifestyle, appliance usage, and economic activities. These variations can make it difficult to identify universal patterns and predict future demand accurately.

The project may also require advanced technologies such as smart meters, data analytics tools, and monitoring systems, which can be costly to implement and maintain. Additionally, collecting detailed consumption data may raise privacy concerns among users, as energy usage information can reveal personal habits and routines.

Furthermore, the findings and recommendations may not always be applicable to all regions or consumer groups due to differences in infrastructure, climate, energy policies, and socioeconomic conditions. Changes in technology and consumer behaviour can also quickly alter electricity consumption trends, requiring continuous updates and monitoring.

Key Disadvantages

- Dependence on accurate and complete electricity consumption data.
- High cost of data collection, monitoring, and analysis technologies.
- Difficulty in predicting consumption patterns due to changing user behaviour.
- Privacy and security concerns related to energy usage data.
- Variations in consumption across different regions and consumer groups.
- Time-consuming process of collecting, cleaning, and analysing large datasets.